

Math 241, F14
Quiz 8

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 3 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) If $z = f(x, y) = ye^{xy}$, find an approximation of $f(1.1, -0.05)$ by using the linear approximation for f at $(1, 0)$.

$$\begin{cases} f_x = y^2 e^{xy} \\ f_y = e^{xy} + xy e^{xy} = (1+xy)e^{xy} \end{cases} \Rightarrow \begin{cases} f_x(1, 0) = 0 \\ f_y(1, 0) = 1 \end{cases}$$

$$\begin{aligned} f(x, y) &\approx f(1, 0) + f_x(1, 0)(x-1) + f_y(1, 0)(y-0) \\ &= 0 + 0 \cdot (x-1) + 1 \cdot y \\ &= y \end{aligned}$$

$$\text{Thus } f(1.1, -0.05) \approx -0.05.$$

Problem 2. (5 points) Find the directional derivative of the function $f(x, y) = \sin(xy)$ at the point $(1, 0)$ along the direction of the vector $\mathbf{v} = \langle 1, 2 \rangle$. What is the maximum rate of change of f at this point?

$$\begin{aligned} f_x &= y \cos(xy), \quad f_y = x \cos(xy) \\ \Rightarrow f_x(1, 0) &= 0, \quad f_y(1, 0) = 1, \quad \text{i.e. } \nabla f(1, 0) = \langle 0, 1 \rangle \end{aligned}$$

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\langle 1, 2 \rangle}{\sqrt{1^2 + 2^2}} = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$$

$$\Rightarrow D_{\mathbf{u}} f = \nabla f(1, 0) \cdot \mathbf{u} = \langle 0, 1 \rangle \cdot \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle = 0 \cdot \frac{1}{\sqrt{5}} + 1 \cdot \frac{2}{\sqrt{5}} = \frac{2}{\sqrt{5}}$$

Maximum rate of change of f at $(1, 0)$ is given by

$$|\nabla f(1, 0)| = |\langle 0, 1 \rangle| = 1$$